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AI Driven Autonomous Gas-Lift Optimization: Enabling Production Enhancement Through Digital Transformation

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Abstract

This paper presents a pioneering approach to autonomous gas-lift optimization using AI-enabled Multivariable Predictive Control (MPC), aligned with the upstream oil and gas sector's digital transformation strategy. The solution integrates data-driven and machine learning models to enable real-time adaptation and optimization of oil-producing wells, targeting enhanced production with reduced gas-lift energy consumption. Automated workflows support cross-functional collaboration and streamline operations.

Dynamic linear well models were developed using step test data from selected wells. A clustering technique based on choke orifice, productivity index among other parameters (Hernandez et al., 2024 [1]) enabled model parameter generalization across similar wells, minimizing the need for individual step tests. Inferential models were deployed to estimate bottom-hole pressure and oil rate in the absence of direct measurements. Advance Process Control (APC) strategies were implemented on a centralized Level 3 control server, integrated with the Distributed Control System (DCS), while cloud-hosted workflows managed model updates and operating envelope adjustments.

The system-maintained gas-lift at the inflection point or up to a customizable economic limit (GAIN FACTOR) in terms of BLS per MMSCF, achieving maximum energy efficiency and up to 1–2% increase in overall production. A supervisory cloud layer continuously tracked soft sensor outputs and field controller data, triggering automatic recalibration workflows when deviations occurred. Digital dashboards and decision support tools provided real-time insights and well-specific recommendations, enhancing operational agility and reducing downtime.

This study validates the potential of AI-driven optimization to automate complex field operations, reduce human intervention, and enable scalable, closed-loop control across gas-lifted assets demonstrating a tangible pathway for digital transformation in upstream oil and gas operations.

Introduction

The upstream oil and gas industry is undergoing significant transformation, driven by the imperative to improve operational efficiency, reduce energy consumption, and maximize hydrocarbon recovery. Central to this evolution is the adoption of digital technologies that enable intelligent, data driven decision-making and automation of traditional manual processes. Among these, gas-lift optimization stands out as a critical area where digital innovation can deliver substantial production gains.

Historically, gas-lift operations have relied on steady-state assumptions, manual tuning, and field experience to manage well performance. These conventional methods are often reactive, labor-intensive, and unable to adapt dynamically to changing reservoir conditions, equipment behavior, or operational constraints. As a result, opportunities for optimization are frequently missed, and production potential remains underutilized.

The emergence of AI-enabled control systems offers a paradigm shift in how gas-lifted wells are managed. By leveraging continuous data streams, predictive analytics, and machine learning algorithms, it is now possible to implement real-time, autonomous optimization strategies that respond proactively to field conditions. This approach not only enhances production but also ensures alignment with safety, integrity, and energy efficiency goals.

This Well Autonomous Control System (AWC) initiative exemplifies this transformation. It introduces a scalable, adaptive, and intelligent framework for autonomous well operations, integrating multivariable predictive control (MPC), inferential modeling, and cloud-hosted supervisory workflows. The system operates within a secure dual-network architecture, interfacing with field-level Distributed Control Systems (DCS) and enterprise-level digital platforms to deliver closed-loop optimization. This paper presents the development and large-scale deployment of an AI-driven, autonomous gas-lift optimization system deployed across hundreds of wells. It highlights the modeling strategies that served as a catalyst for digital transformation, enabling scalable control and inferential estimation in data-limited environments. The work further describes how the modeling framework was operationalized within a digital operating model, integrating hybrid control logic into both IT and OT environments to support real-time optimization and autonomous execution.

Detailed user interfaces are presented, showcasing how visualization tools were designed to empower petroleum engineers and operators with actionable insights and situational awareness. Finally, the study offers a structured roadmap for digital transformation in upstream operations, outlining key enablers such as model-driven intelligence, workflow automation, and cross-functional collaboration.

Artificial Intelligence Techniques for Gas Lift network modelling & Optimization

The complexity and non-linearity of gas lift systems present significant challenges for traditional optimization methods, which often rely on steady-state assumptions and are computationally intensive. To address these limitations, artificial intelligence (AI) techniques have emerged as powerful tools for modeling and optimizing gas lift networks in real time.

Machine learning approaches, particularly Artificial Neural Networks (ANNs), can capture nonlinear relationships between operational variables by learning from historical data. While they offer strong predictive capabilities, their effectiveness depends on the availability of large, high-quality datasets and substantial computational resources. Additionally, their "black box" nature can limit interpretability, which is critical for field validation and operator trust.

Meta-heuristic algorithms such as Genetic Algorithms (GAs) provide flexible optimization strategies that can navigate complex solution spaces. However, they are often computationally expensive and may suffer from convergence issues. Fuzzy logic systems, which excel in handling uncertainty and imprecise inputs, are well-suited for modeling ambiguous gas lift conditions but require expert-defined rules and tuning.

Response Surface Methodology (RSM), a statistical technique, supports structured experimentation and sensitivity analysis of operational parameters. While effective in controlled environments, RSM assumes well-defined variable relationships and may not scale efficiently in dynamic field conditions.

To overcome the individual limitations of these techniques, hybrid AI models are gaining traction. These models combine the predictive strength of machine learning, the adaptability of meta-heuristics, the uncertainty management of fuzzy logic, and the analytical rigor of statistical methods.

Deployed Artificial Intelligence Technique for Well & Gas Lift modelling and Optimization

All control-critical estimates are delivered by RSM based inferential models deployed on-premises within the field control room and fully integrated with the APC system. In parallel, a cloud-hosted data-driven Virtual Flow Meter (VFM) provides production estimation and reconciliation ensuring that the on-premises RSM models remain accurate, validated, and operate within defined guardrails.

Among the various artificial intelligence and optimization methods summarized in Table 1, this study employs Response Surface Methodology (RSM) and Artificial Neural Networks (ANNs). RSM is selected for its computational efficiency, stability, and interpretability in developing surrogate models for inferential estimation, while ANNs are leveraged for their ability to capture complex nonlinear dependencies in real-time production data, complementing the physics-driven surrogate framework.

Table 1—Summary of Artificial Intelligence Techniques for Gas Lift network Modelling & Optimization [2]

Type of Optimization Method	Principle	Application for Gas Lift Optimization	Advantage	Limitation
Artificial Neural Networks (ANN)	Mimics the human brain's network of neurons to model complex patterns and relationships in data.	Used to predict production rates based on historical data, optimizing gas injection rates.	Can model complex, non-linear relationships; adaptable to changing data.	Requires large datasets and significant computational resources; can be a "black box" model, difficult to interpret.
Genetic Algorithm (GA)	Uses principles of natural selection and genetics to evolve solutions to optimization problems over generations.	Optimizes gas allocation by evolving a population of solutions, balancing gas injection rates across wells.	Explores a diverse set of solutions; good for complex, multi-modal problems.	Computationally expensive, may converge with local optima; requires careful parameter tuning
Fuzzy Logic	Utilizes degrees of truth rather than binary logic to handle uncertainty and approximate reasoning.	Models uncertain and imprecise information in gas lift systems, optimizing operational parameters.	Handles uncertainty well; provides interpretable rules and reasoning.	Can be less precise; requires expert knowledge to define membership functions and rules.
Response Surface Methodology (RSM)	Uses statistical and mathematical techniques to model and analyse problems where several variables influence the outcome.	Develops predictive models of gas lift performance, optimizing operational conditions through designed experiments.	Efficient exploration of parameter space; provides statistical validation of results.	Assumes well-defined relationships between variables; can be resource-intensive to design and execute experiments.

Inferential Models RSM-based Methodology

All inferential models are implemented as Well Model-proxy RSMs, they are fitted using Well Model-generated sensitivity data, reproducing the calibrated well model's behavior across the operating window, but running locally at control frequency without Well Model licensing or runtime overhead. The formulation assumes the well is operating under steady-state conditions, so it will underestimate flush production, which can last for several days in tight reservoirs. The inferential models developed for this study include:

Bottom Hole Pressure Inferential Model (BHPIM): provides real time estimates of bottomhole flowing pressure (BHFP). In practice, each asset defines a minimum BHFP threshold based on reservoir management guidelines to safeguard pressure support and long-term recovery. This minimum value is enforced as constraint in the APC optimization loop. The formulation is as follows:

$$PWF = f(WHP, GL, BSW, GOR, P, RES, PI)$$

Production Rate Inferential Model (PRIM): estimates steady state oil production based on gas injection and other surface/reservoir inputs. For oil, a per well production target is typically assigned by asset teams, which the optimizer seeks to achieve or maximize subject to gas lift and reservoir-operational constraints. Thus, oil rate acts as a control objective within the optimization framework. The formulation is as follows:

$$OIL = f(WHP, GL, BSW, GOR, P, RES, PI)$$

Gas Lift Production Ratio Inferential Model (GLPRIM): represents the marginal oil gain per unit of injected gas. Derived analytically from PRIM, this metric underpins the determination of per-well optimum gas lift rates. The formulation is as follows:

$$GLPRIM = \frac{\partial \widehat{Q}_o}{\partial GL}$$

Input to these models includes wellhead pressure and temperature, reservoir pressure, gas lift injection rate, gas oil ratio (GOR), water cut (WCUT), and productivity index (PI). These parameters are embedded into polynomial regression equations that approximate multiphase flow behaviour while maintaining transparency and control interpretability.

The formulation results in 28 coefficients for OIL and 28 coefficients for PWF to account for parameter interactions. The polynomial is of the form:

$$\begin{aligned} f(x_1, x_2, x_3, x_4, x_5, x_6) = & a_0 + a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + \\ & a_7x_1^2 + a_8x_2^2 + a_9x_3^2 + a_{10}x_4^2 + a_{11}x_5^2 + a_{12}x_6^2 + a_{13}x_1x_2 + a_{14}x_1x_3 + a_{15}x_1x_4 + \\ & a_{16}x_1x_5 + a_{17}x_1x_6 + a_{18}x_2x_3 + a_{19}x_2x_4 + a_{20}x_2x_5 + a_{21}x_2x_6 + a_{22}x_3x_4 + \\ & a_{23}x_3x_5 + a_{24}x_3x_6 + a_{25}x_4x_5 + a_{26}x_4x_6 + a_{27}x_5x_6 \end{aligned}$$

Note these proxy models are fitted over a specific range of values. At the moment, this requires relatively frequent bulk updates of the coefficients if Well model IPR parameters are updated, or the APC high/low limits are updated to significantly different values.

The complete workflow for developing and updating the inferential models is shown in the [Figure-1](#) below, which illustrates the sequence from well model verification, sensitivity setup, and scenario simulation, through to RSM construction and applicability checks.

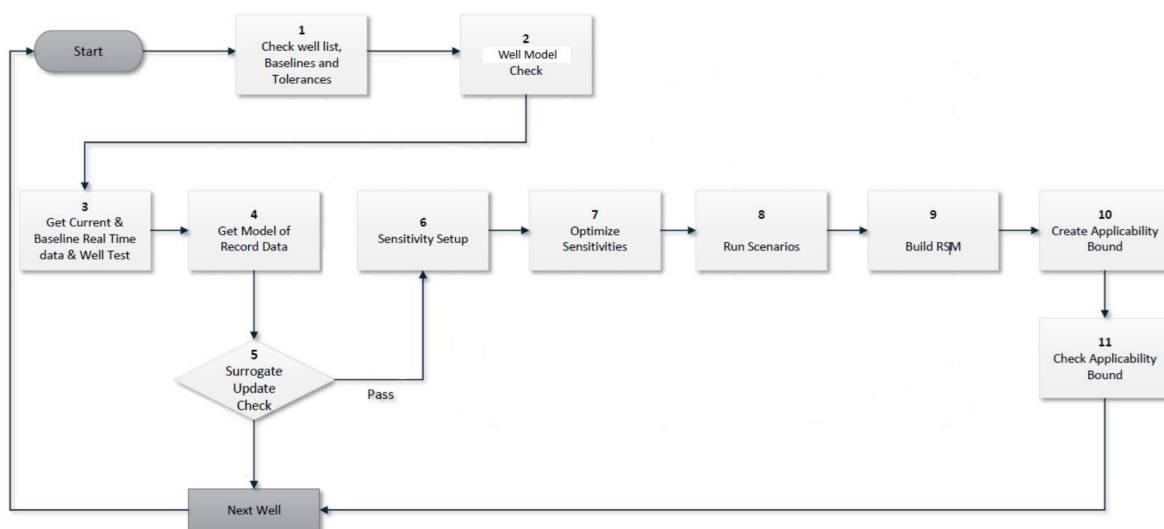


Figure 1—Workflow for RSM-based inferential model development

Figure 2 illustrates the performance of the RSM-based inferential oil rate model compared against the calibrated well model and measured wellhead flowing pressure (WHFP) over a two-week period. The inferential oil rate closely follows the reference well model oil rate, with deviations consistently within $\pm 4\%$ error.



Figure 2—Example of Inferential Well Model Performance

The observed variations in WHFP and gas-lift injection (GL) during the monitoring period are accurately captured by the inferential, demonstrating its ability to reproduce the dynamic response of the well under changing operating conditions. The low error margin confirms that the surrogate (Well Model-proxy RSM) provides a computationally efficient yet reliable representation of the calibrated well model, suitable for deployment in high-frequency APC loops.

Gas-Lift Optimization logic

The computation of optimum gas-lift injection is governed by a "Gain factor", corresponding to the slope of the gas-lift performance curve (GL vs. Oil). A default slope of 50 STB/D per MMSCFD is applied, but this value can be adjusted by the I&C engineer at the APC engineering station to reflect economic conditions.

Crucially, the optimizer does not target the absolute maximum of the GL curve. Instead, it selects the operating point where incremental injection no longer yields meaningful economic benefit, expressed as:

$$\text{Optimum GL} = GL \left| \frac{\partial \widehat{Q_o}}{\partial GL} \right|_{WHP} = \text{Gain Factor}$$

If the PRIM model is quadratic with interaction terms, the optimum gas-lift can be computed directly, since the derivative of a quadratic function is linear:

$$\frac{\partial f}{\partial x_1} = a_1 + 2a_7x_1 + a_{13}x_2 + a_{14}x_3 + a_{15}x_4 + a_{16}x_5 + a_{17}x_6$$

where x_1 corresponds to gas-lift injection rate, and the remaining x_i capture reservoir and operational parameters (e.g., WHP, WC, GOR, PI). The optimization cycle executes every one minute, recomputing the GL optimum based on the updated WHP. To mitigate oscillations, a deadband of ± 0.05 MMSCFD is applied. Over successive cycles, the APC reads the current WHP, updates the GL target, and accounts for backpressure feedback. As WHP stabilizes, the GL setpoint converges.

This closed-loop methodology implicitly incorporates cross-well interactions and backpressure impacts, functioning similarly to GAP's rules-based solver with equal-return allocation. The key distinction is that

the APC operates on real-time field measurements, ensuring convergence to stable, economically optimal setpoints under actual operating conditions.

Artificial Neural Networks (ANNs) Modelling

To complement the RSM-based inferentials, Artificial Neural Networks (ANNs) are deployed as data-driven Virtual Flow Meters (VFM) capable of predicting well production behaviour directly from surface sensor data. The ANN predictions are continuously compared with RSM outputs (PRIM) to detect deviations, thereby enhancing reliability and enabling adaptive optimization.

Methodology. ANNs are introduced to overcome the limitations of parametric surrogate models. Unlike RSMs, which approximate physical behaviour with polynomial equations, ANNs can learn nonlinear relationships directly from data, providing robust predictions under variable reservoir and operating conditions.

Input Features and Data Preparation. The input data is based on a comprehensive dataset comprising real-time sensor and historical production data. Input features include wellhead pressure and temperature, bottomhole pressure and temperature (measured or estimated), choke opening, flowline and manifold pressures and temperatures, gas-lift injection rate and temperature, casing head pressure, and compositional indicators such as Gas-Oil Ratio (GOR) and Water Cut (WC). These high-dimensional inputs are standardized and processed using time-series windowing techniques to capture temporal dependencies and production trends. Data is retrieved from Corporate Plant historians and made available in the Cloud through level 4 secure APIs. Prior to training, a rigorous pre-processing workflow is applied to improve model fidelity by removing noise and non-representative cases. This includes the exclusion of unstable MPFM readings at the start and end of tests (± 10 minutes), Z-score filtering ($|Z| > 2$), and the removal of values outside the 5th–95th percentile range. Additional rules enforce single-well routing during MPFM testing (discarding cases where multiple wells are commingled) and resampling of the training dataset from 1-minute to 15-minute intervals to reduce noise and improve generalization.

Model Training and Validation. The ANN is trained in a supervised regression framework to predict oil, gas, water and liquid production. To ensure temporal consistency and realistic performance evaluation, a Time Series Split cross-validation approach with five sequential folds was employed, preserving the chronological order of the data so that future predictions were always based on past information. Model accuracy was assessed using the Median Mean Absolute Percentage Error (MAPE), selected for its robustness to extreme outliers compared with the traditional mean MAPE. The core model architecture was a feed-forward ANN optimized via backpropagation, and its performance was systematically benchmarked against ensemble tree-based methods (e.g., ExtraTrees) to validate predictive robustness and minimize the risk of overfitting.

Operational Deployment and Cloud Integration. The ANN model is deployed within the AIQ Cloud environment, leveraging the same digital infrastructure that hosts the RSM-based inferentials to ensure seamless integration across the optimization framework. Once trained, the model operates in real time, generating production rate predictions at a one-minute execution frequency from streaming field sensor data. Model integrity is maintained through automated retraining routines and drift detection mechanisms, which continuously benchmark predictions against historical logs and update model parameters as required. The deployment is further enhanced by integration with visualization dashboards and alerting systems, enabling engineers to monitor performance and respond proactively to deviations. To strengthen reliability, a hybrid inference strategy is implemented in which ANN predictions are cross validated against RSM outputs, providing redundancy and improving confidence in production estimates used for gas lift optimization.

The figure below shows the Median MAPE per well, for wells from different clusters.

ENSEMBLE MODEL MAPE

Cluster	Well Name	Config	Start Date	End Date	MAPE
CLS		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.91%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'CHP_A', 'FLP]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.89%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'CHP_A', 'FLP]	09/07/2021 1:01:00 PM	14/12/2022 9:45:00 AM	1.02%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'CHP_A', 'FLP]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.77%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'CHP_A', 'FLP]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.41%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.17%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.18%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	2.80%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.45%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	11.23%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.25%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.65%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.27%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	4.47%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.67%
CLS		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.35%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.47%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	3.99%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.44%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.97%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	4.22%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.20%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT', 'BHP_A', 'BHT_A']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.02%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'CHP_A', 'FLP]	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.20%
		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	0.49%
CLS		_ST [FLT, 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.83%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:01:00 PM	14/12/2022 9:45:00 AM	1.92%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	2.66%
CLS		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	2.02%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.94%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	1.50%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	7.17%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	25.35%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/07/2021 1:00:00 PM	14/12/2022 9:45:00 AM	2.25%
		_ST [FLT, 'MT', 'WHP', 'WHT', 'CHK_PCT', 'GL', 'CHP_A', 'MP', 'FLP', 'GL_CHT']	09/09/2021 12:15:00 AM	14/12/2022 9:45:00 AM	10.76%

Figure 3—Model MAPE for wells from different clusters.

Modelling as a Catalyst for Digital Transformation

In the context of upstream digital transformation, the use of models extends far beyond engineering calculations. It becomes a foundational layer for automation, cross-functional integration, and scalable deployment of intelligence across assets. Rather than viewing modeling as an isolated technical task, this work frames it as a strategic enabler driving the transition from manual, reactive operations to fully autonomous well control.

Building on the modeling framework previously established in Hernandez et al. (2024) [1], this study highlights how dynamic linear and inferential models were operationalized to support a digital operating model. These models, trained on historical and step test data, were not only used to control individual wells but served as reusable, auditable building blocks to standardize optimization logic across the field. Their integration into Advanced Process Control (APC) systems reduced the dependency on expert tuning, allowing operations teams to focus on decision-making rather than manual intervention.

Key enablers of this digital transformation included:

- **Model-Driven Intelligence:** Use of real-time inferential models to estimate production parameters that were previously unavailable without costly instrumentation.
- **Template-Based Scalability:** Deployment of control models across well clusters using structured parameterization, reducing engineering workload and ensuring consistency.
- **Automation of Engineering Workflows:** Embedded bias correction, model reparameterization, and bulk updates replaced repetitive manual tasks, accelerating responsiveness to field changes.
- **Bridging OT and Data Science:** Hybrid models served as a convergence point for control engineers, data scientists, and production teams, fostering a unified approach to optimization.

By positioning modeling not just as a technical enabler but as a cornerstone of digital operating models, this approach unlocks a pathway for organizations to scale autonomy, enhance agility, and future-proof production systems in increasingly complex operational environments.

The next section outlines how this modeling framework was deployed within a digital operating model, addressing scalability, integration, and collaboration challenges necessary to sustain autonomous optimization in a live production environment.

Enabling the Digital Operating Model

Translating a modeling framework into sustainable field impact requires more than technical capability, it demands a robust digital operating model. This model must ensure that intelligence generated from dynamic and inferential models is reliably deployed, integrated with field systems, and maintained across a growing set of wells. The successful deployment of AWC involved a multi-layered strategy addressing four key pillars: scalability, system integration, operational governance, and cross-functional collaboration.

Scalable Model Deployment and Maintenance

To scale model-driven control across the field, a parameterized template was adopted. Dynamic linear models and inferential models—originally developed on representative wells—were encoded into reusable templates. These templates captured core model structures and control logic while allowing well-specific parameters (e.g., productivity index, reservoir pressure, gain-factor, coefficients, inferential constants) to be customized via structured data inputs (CSV file).

A centralized deployment pipeline enabled:

- Bulk propagation of model coefficients to the APC system,
- Version control for model updates,
- This system allowed field engineers to onboard new wells rapidly and ensured consistency in the applied optimization logic.

Integration with OT and Control Infrastructure

Autonomous optimization required integration with the existing OT layer, including Distributed Control Systems (DCS), safety logic, and SCADA infrastructure. The deployment architecture adopted a hybrid setup, where real time data acquisition and actuation were managed at the edge via DCS and field instruments, while well model recalibrations via cloud connected layer models & is executed by APC. To safeguard operations, watchdog logic and fallback mechanisms were implemented to detect:

- Sensor failures,
- Communication losses, and
- Invalid model predictions.

These safeguards ensured that control actions remained within defined safety limits and that manual override was always possible.

Digital Workflows and Model Lifecycle Governance

Maintaining model accuracy over time is essential for sustainable autonomy. To this end, digital workflows were established to manage the full lifecycle of deployed models. These included:

- Bias correction loops that compared predicted vs. measured values and applied systematic adjustments.
- Trigger-based reparameterization, activated when operating conditions drifted outside model validity ranges.

- Bulk parameter update workflows, enabling efficient propagation of revised inferential and control model parameters.

These workflows minimized manual intervention and ensured that the system remained responsive to evolving dynamics.

Cross-Functional Collaboration and Cultural Shift

One of the most critical components of operationalizing this model was driving alignment across disciplines. The project brought together:

- Production engineers, responsible for optimization logic and performance targets,
- Control engineers, overseeing OT integration and regulatory compliance, and
- Data scientists, managing model training, clustering, and validation.

By formalizing shared workflows, creating joint ownership of model outputs, and using intuitive dashboards for monitoring, the initiative bridged long-standing silos. This collaboration enabled the system to transition from a pilot deployment to a fully integrated field-wide solution.

Digital Interfaces for Transparency and Control

To enable seamless collaboration, improve operational transparency, and ensure wide user adoption, a suite of digital interfaces was developed across both the IT (enterprise) and OT (field) layers. These interfaces were designed not only to monitor the status of autonomous operations but also to digitally bridge the communication gap between petroleum engineers and field operators, eliminating the dependency on unstructured, verbal instructions.

IT Layer Dashboards – Centralized Monitoring and Control Configuration

The IT-based engineering dashboards serve two primary purposes:

- Surveillance and performance oversight offers engineers a consolidated, real-time view of well behavior, control actions, and deviations from expected performance.
- Operational task management allows engineers to log required actions, define limits and targets, and assign executable tasks to the OT layer through a structured, traceable workflow.

Together, these interfaces represent a critical step toward full autonomy by supporting situational awareness, model validation, and collaborative decision-making within a secure, digital environment. The following key user interfaces were developed:

Well Operation Requirements. To eliminate reliance on informal communication channels, particularly phone calls between petroleum engineers and field operators—a dedicated Well Requirements Interface was developed within the IT environment. This interface enables engineers to digitally log operational requests, including updated production targets, control activation instructions, and revised operating limits for each well. Once submitted, the system automatically generates corresponding tasks, which are assigned to the relevant operators within the OT environment.

This structured approach addresses a critical integration gap: the absence of direct, secure communication from the IT layer to the OT layer. Historically, instructions were conveyed verbally, leading to a lack of traceability, inconsistent implementation, and increased operational risk. By digitizing this process, the Well Requirements Interface ensures that control logic changes are properly documented, transparently assigned, and consistently executed thereby enhancing accountability, minimizing miscommunication, and reinforcing operational discipline in autonomous well control workflows.

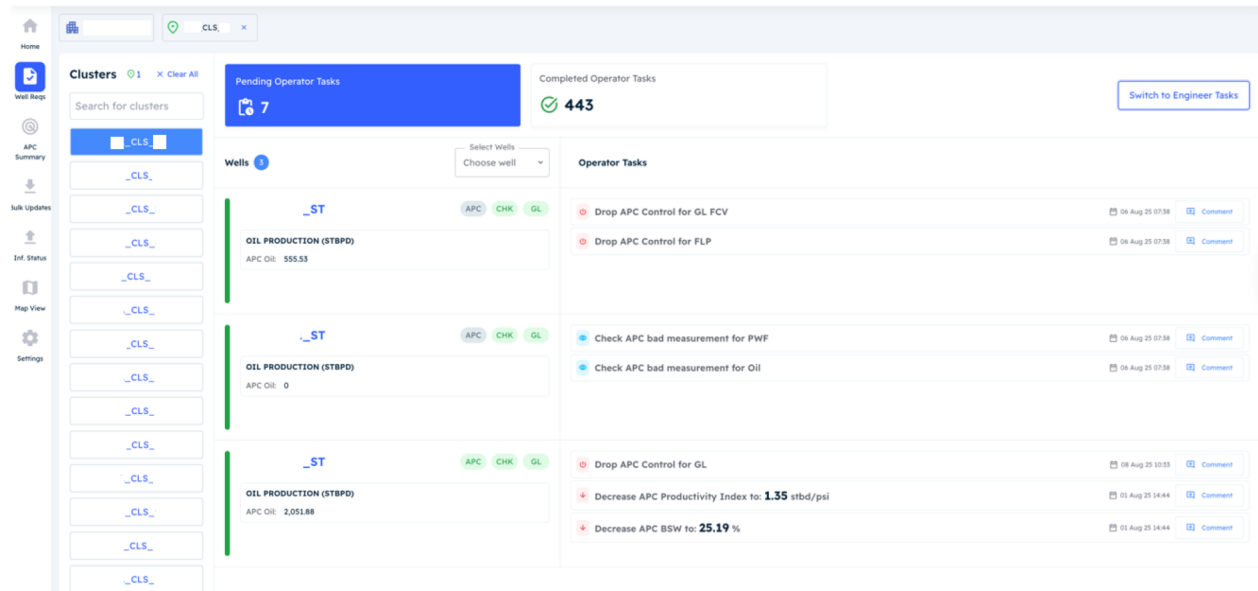


Figure 4—AWC Well Requirements User Interface

Radar Plot (Spider Chart) – Operator View. This view provides a graphical representation of live measured values against defined high and low limits for key control variables. It helps operators instantly identify whether the well is operating within its envelope, flagging any deviations from the control strategy.

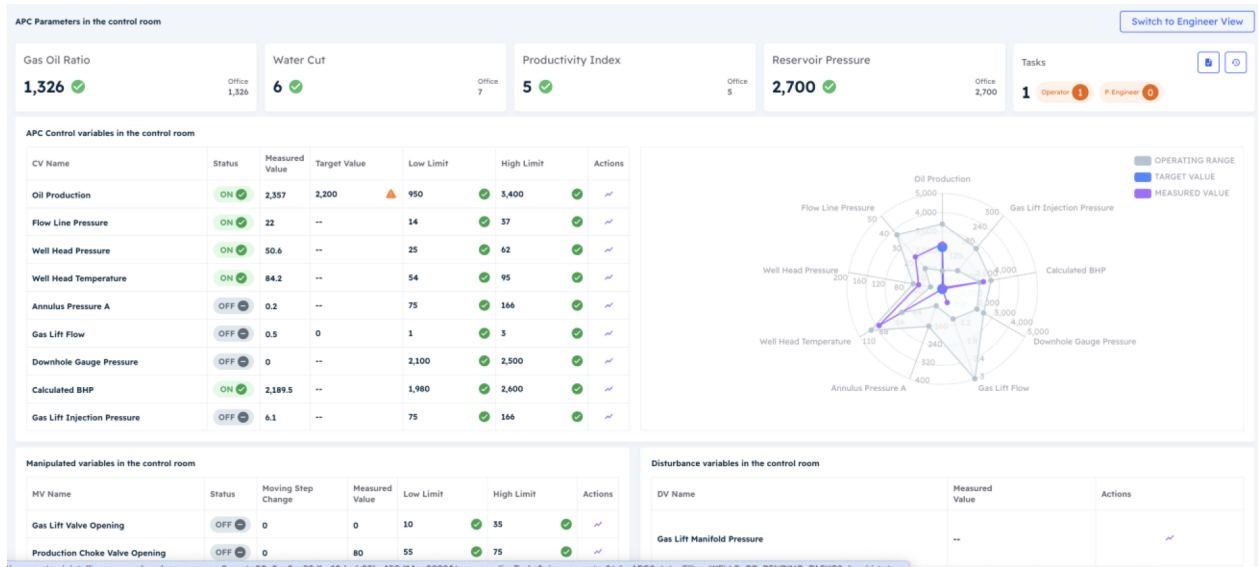


Figure 5—AWC Operating Envelope (Operator View) User Interface

Radar Plot (Spider Chart) – Engineer View. A parameter configuration panel that allows engineers to set control targets, operational boundaries (min/max), and indicate which control loops should be activated. Once updated, associated tasks are automatically created and assigned to the operations team for execution.

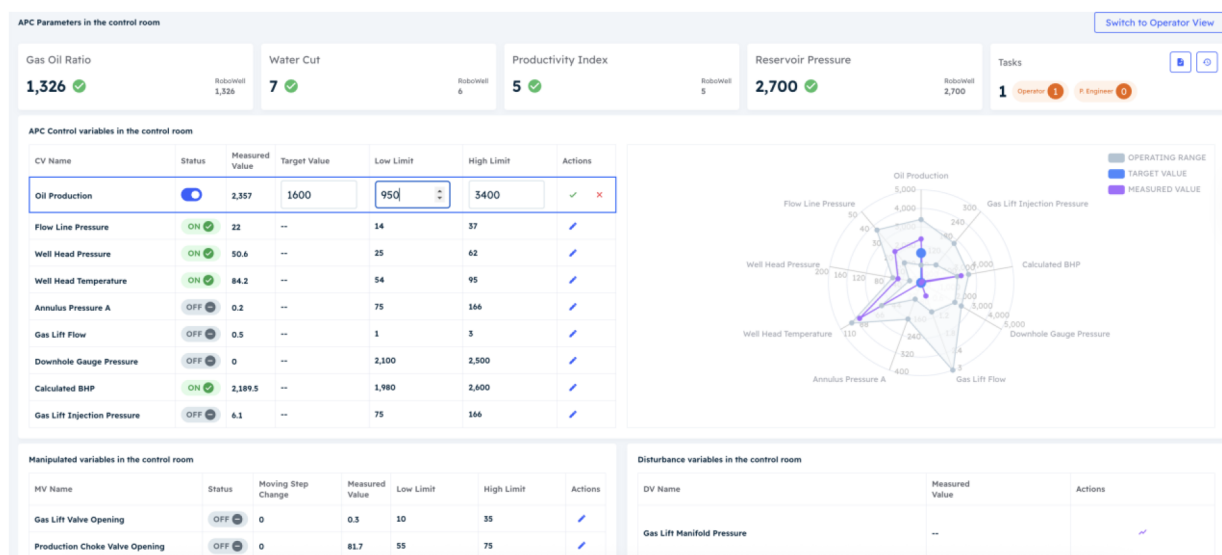


Figure 6—AWC Operating Envelope (PE View) User Interface

APC Status Dashboard. A real-time monitoring interface showing the status of manipulated and controlled variables including their control mode (ON, OFF, FREEZE) and signal health (GOOD, BAD). Colored overlays enable engineers to quickly assess loop performance and data integrity over time.

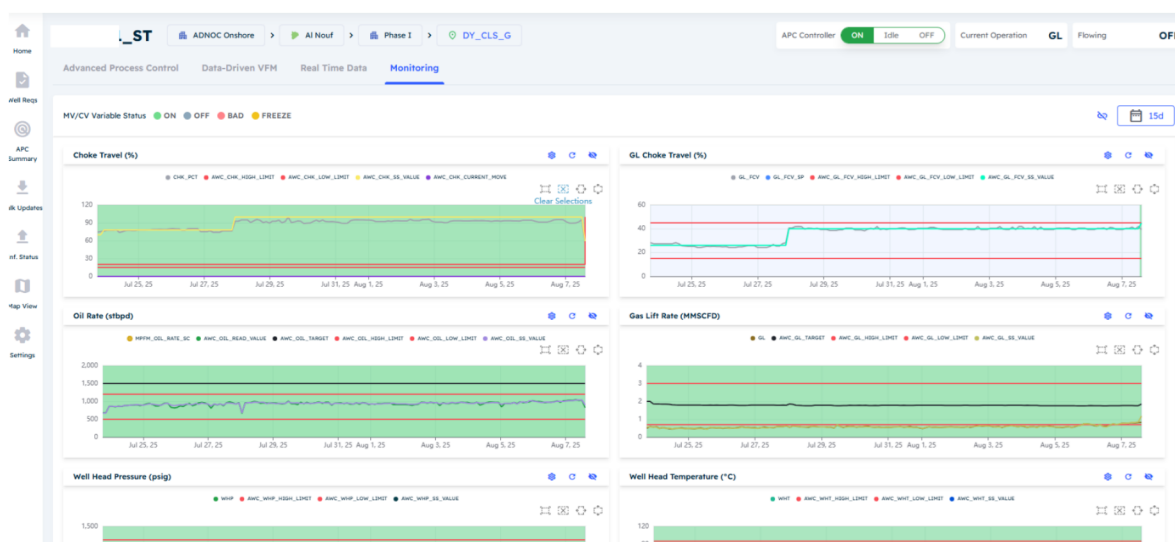


Figure 7—AWC Controller Status User Interface

Homepage KPI Summary. A centralized top-level interface designed to provide a real-time overview of field-wide performance and operational compliance. This dashboard displays key indicators such as the total number of wells, number of flowing wells, wells with AWC enabled or disabled, and a list of flagged wells—specifically those that are producing but have autonomous control turned off, or those exhibiting configuration discrepancies.

In addition, the interface monitors constraint deviations across the well population and provides actionable summaries of outstanding tasks assigned to both operators and petroleum engineers. These include control actions such as enabling APC, updating limits, or reviewing model behavior. At the field level, the dashboard also presents aggregated oil production versus target, and gas lift injection versus target, enabling engineers to quickly assess optimization performance and take corrective measures where needed.

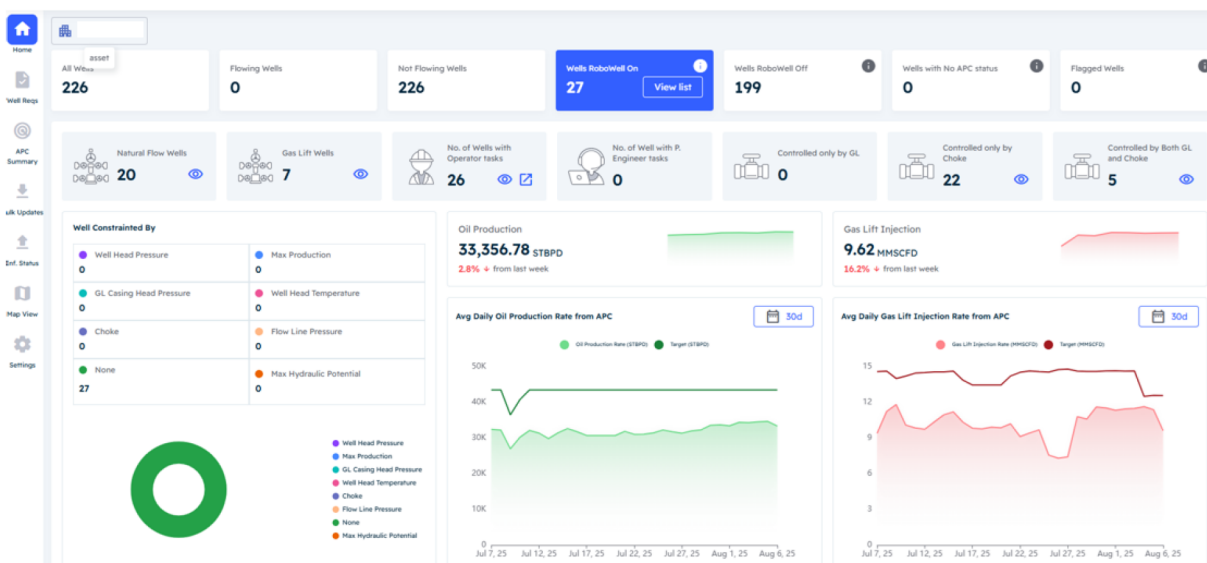


Figure 8—AWC Homepage User Interface

By integrating these interfaces into a unified digital workspace:

- Petroleum engineers gain full visibility and control over autonomous systems without field intervention.
- Operators receive clear, actionable tasks with traceable context, improving execution efficiency.
- The organization benefits from reduced miscommunication, greater operational discipline, and a shared understanding of system behavior across domains.

OT Layer Interfaces – Enabling Field Adoption and Operational Simplicity

To complement the engineering dashboards developed in the IT environment, user-friendly interfaces were also implemented in the OT layer, specifically within the Distributed Control System (DCS). These field-facing screens were designed to promote user adoption among control room operators and to ensure that well AWC could be easily understood, trusted, and operated at the wellsite.

The OT interfaces featured:

- Simplified visualizations of key process variables
- Real-time feedback on control loop activity (e.g., choke and gas lift valve positions)
- Alarm states and limit violations
- Manual override options when required

By aligning with existing operator workflows and using familiar DCS screen designs, these interfaces minimized resistance to adoption and ensured that operators could quickly understand and act upon the instructions received through the digital tasking system. They played a critical role in reinforcing the value of autonomous control while maintaining transparency and human-in-the-loop oversight at the field level.

At the control level, the dashboard & DCS Graphics provided real-time visibility into the status of Advanced Process Control (APC) systems. Operators, Engineers and control specialists could monitor:

General APC System Overview providing APC Controller On/Off status and access to deactivate APC control schemes for individual well of the field.

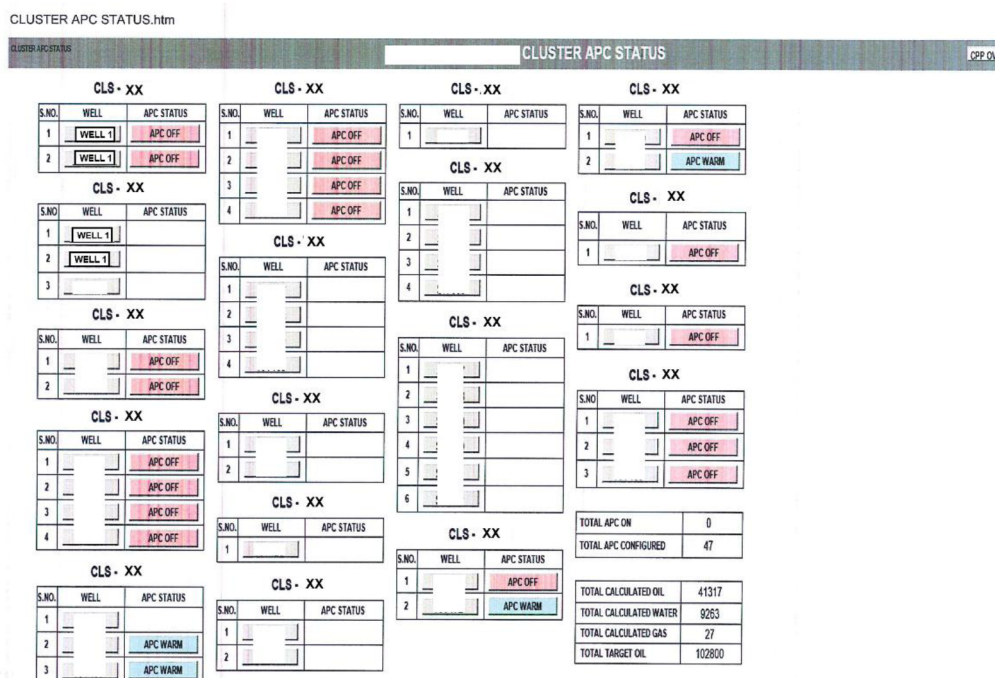


Figure 9—DCS APC Field Overview

DCS Cluster wise Well Summary Screen to visualise key parameters related to APC operations for cluster of wells such as APC on/off status, production and gas lift choke Manipulated Variable (MV) status, target oil production, current oil, gas & water production, oil gain factor (minimum oil gain (BBLs) per gas injected (MMSCFD) and potential oil & gas lift.

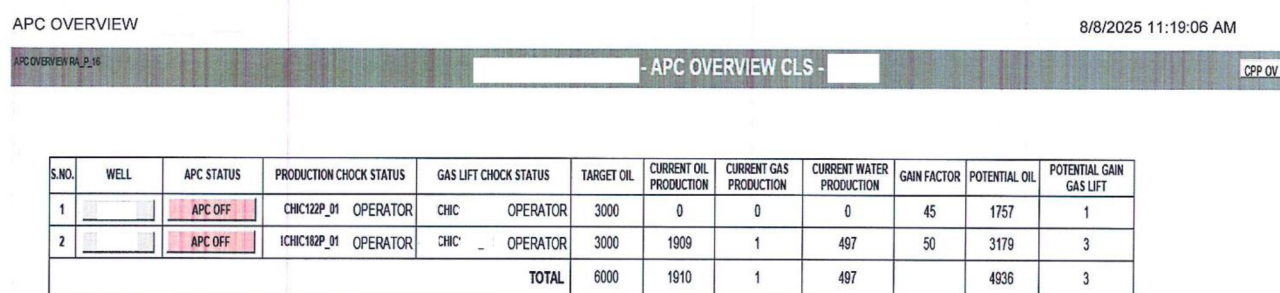


Figure 10—DCS APC Cluster Overview

APC Operating screens for Controlled, Manipulated, and Disturbance Variables (CVs, MVs, DVs), enabling precise tracking of well dynamic behaviour and control action. Deviation alerts and limit violations are automatically flagged when operational parameters exceeded defined thresholds while honouring reservoir process parameters.

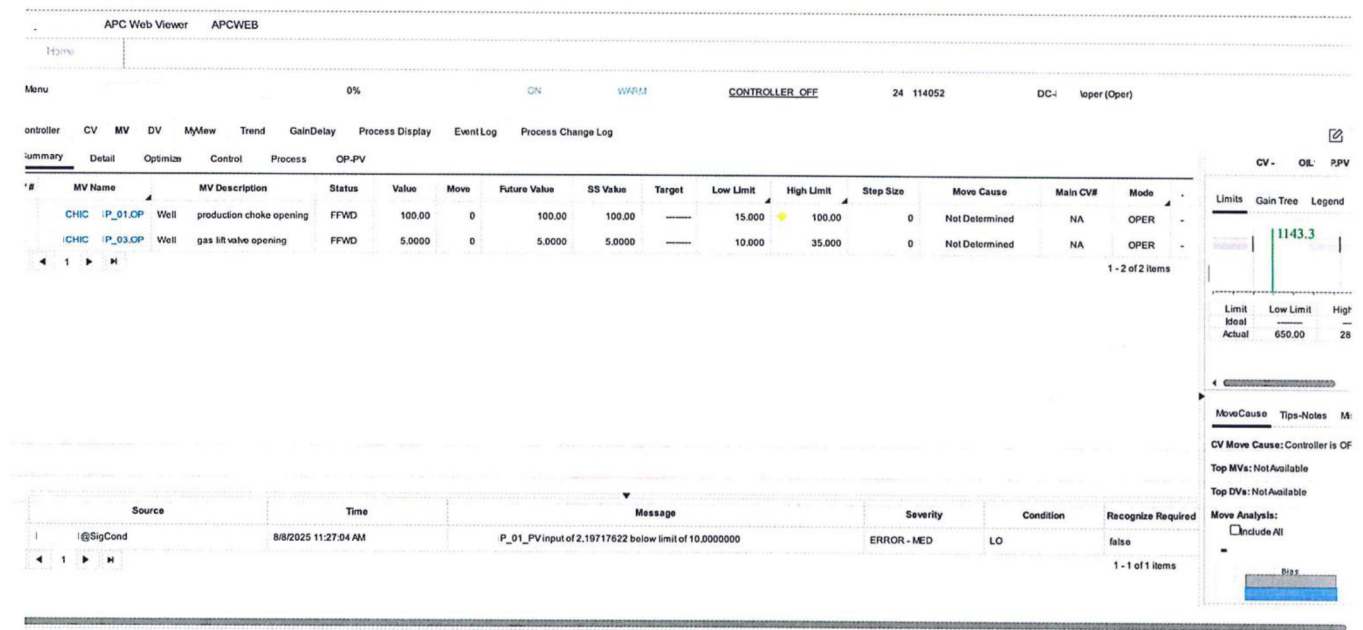


Figure 11—APC Operating Screens – Manipulated Variables

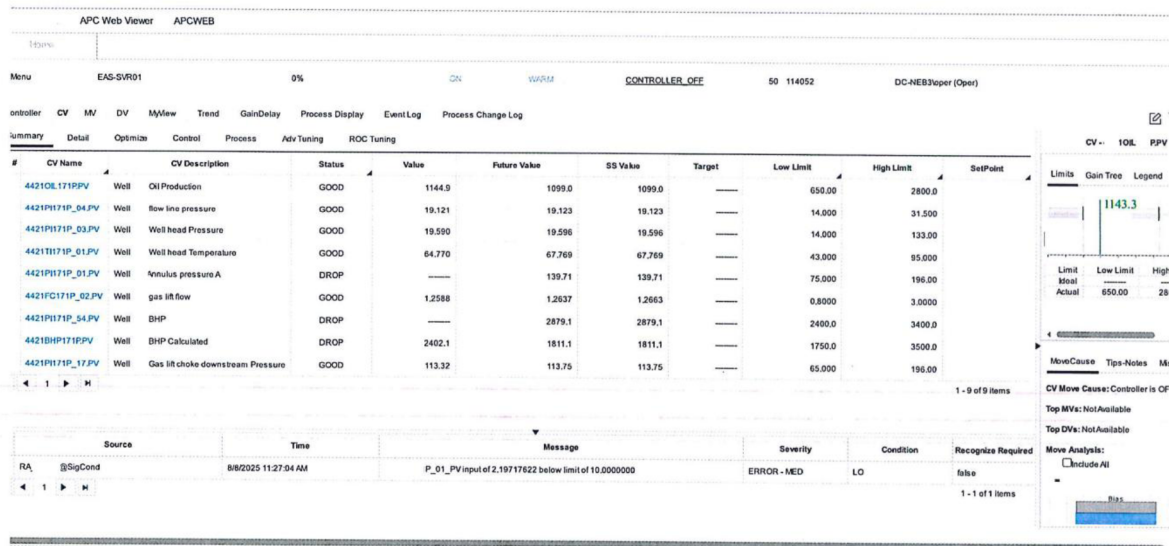


Figure 12—APC Operating Screens – Controlled Variables

Roadmap for Digital Transformation in Gas-Lift Optimization

The transition from conventional, reactive artificial lift management to an autonomous gas lift optimization framework requires a structured and phased approach. This study outlines a practical roadmap that supports upstream operations in navigating the complexities of digital transformation, bridging technology, people and processes.



Figure 13—Roadmap for Digital Transformation in Gas Lift Optimization

Phase 1 – Data Readiness and Diagnostic Assessment

A successful digital transformation begins with evaluating the quality and availability of field data. This includes:

- Auditing well-level instrumentation (pressure, temperature, flow, chokes)
- Assessing historical production and test data integrity.
- Identifying missing or unreliable measurements (e.g., bottomhole pressure, real-time flow rates).
- Classifying wells based on data maturity and automation readiness.

This diagnostic step establishes the foundation for model accuracy, operational trust, and system scalability.

Phase 2 – Model Development and Control Design

In this phase, physics-informed and data-driven models are deployed to infer unmeasured variables and simulate well response.

Key activities include:

- Building dynamic linear models based on step test results and historical data.
- Implementing inferential models to estimate bottomhole pressure and flow rates.
- Using clustering techniques to generalize model parameters across similar wells.
- Designing multivariable predictive controllers (MPC) to adjust key setpoints (e.g., gas lift rate, choke position) in real time.

This step enables predictive, self-adjusting control strategies that replace manual tuning.

Phase 3 – Automation and Field integration

Once models and controllers are validated, the system is integrated into the operational environment:

- Deploying inferential and control logic within the OT layer.
- Ensuring secure connectivity between field devices, control systems, and cloud-based monitoring platforms.
- Introducing watchdog and fail-safe to safeguard production in the event of communication loss, sensor failure, or routing changes.
- Establishing KPIs for optimization performance, production stability and gas utilization efficiency.

This phase enables continuous, closed-loop operation while preserving operational integrity.

Phase 4 – Scaling Strategy

Once validated in a pilot setting, the system must be scaled across additional wells and facilities to realize full value. This requires:

- Developing modular templates for rapid modelling and controller replication.
- Enabling bulk parameter update workflows for efficient deployment of inferential coefficients and control limits.
- Establishing deployment pipelines that allow secure and traceable propagation of updates across multiple assets.

- Integrating the optimization system into enterprise-level surveillance platforms for unified monitoring.

A structured scaling approach ensures consistent performance and minimizes engineering overhead during asset expansion.

Phase 5 – Change Management considerations

Change management is often the most underestimated, yet critical component of digital transformation in upstream operations. Successful adoption requires:

- Early involvement of field operations, control room staff, and petroleum engineers in solution design.
- Transparent communication about the system's role in supporting, not replacing, human decision-making.
- Dedicated training programs and simulation environments to build trust in model behavior and control logic.
- Phased handover of control authority from manual to autonomous modes, with clearly defined fallback procedures.
- Continuous feedback loops incorporate user input and field learning into future iterations.

Failure to address these organizational dynamics can hinder adoption even if the technical solution performs well.

Results, Discussions and Performance Evaluation

The solution enhanced, role-specific visibility transformed conventional siloed workflows. Previously, updates to well models or reservoir assumptions were handled manually and asynchronously. With the new integrated digital ecosystem, model updates automatically interfaced with control systems through cloud-based orchestration. For instance:

- A change in reservoir conditions or constraints like Maximum Allowable Annulus Surface Pressure (MAASP) triggered recalibration workflows for control algorithms
- Soft sensor degradation or changing process dynamics prompted re-training tasks
- APC tuning parameters were adjusted automatically based on model-feedback loops and variance analysis

The result was a closed-loop optimization framework where field operations, digital systems, and engineering models interacted seamlessly, reducing latency in response, improving operational accuracy, and reinforcing the collaboration between field and subsurface disciplines.

Field validation of results was conducted using Well model and real-time wellhead pressure and gas lift data. It involved calculating differential pressure and estimating gas lift savings. The AWC demonstrated its ability to maintain integrity and optimize production without manual intervention.

Use Cases and Operational Scenarios

CASE 1. Prior to the activation of autonomous well control, the gas-lifted well was operated manually, with gas injection and production initiated through operator intervention. During this manual phase, the well exhibited transient behavior introduced disturbances to the downstream processing system. These fluctuations were attributed to inconsistent control actions and delayed response to changing well conditions, which limited production efficiency and compromised process stability.

Following the transition to autonomous operations, the well demonstrated a marked improvement in performance. Production increased by approximately 4–5%, while gas lift injection was optimized by nearly 35%, reflecting more efficient use of lift gas. In addition to these quantitative gains, the well's operating profile stabilized significantly, resulting in smoother flow behavior and reduced variability. This enhanced stability propagated downstream, improving the overall consistency and reliability of the production system. The following sample well trends, i.e., Fig. 14 illustrate the shift from manual to autonomous operation, with green sections on the graphs indicating periods when the AWC controller was active, and light blue sections showing when the AWC controller was inactive.

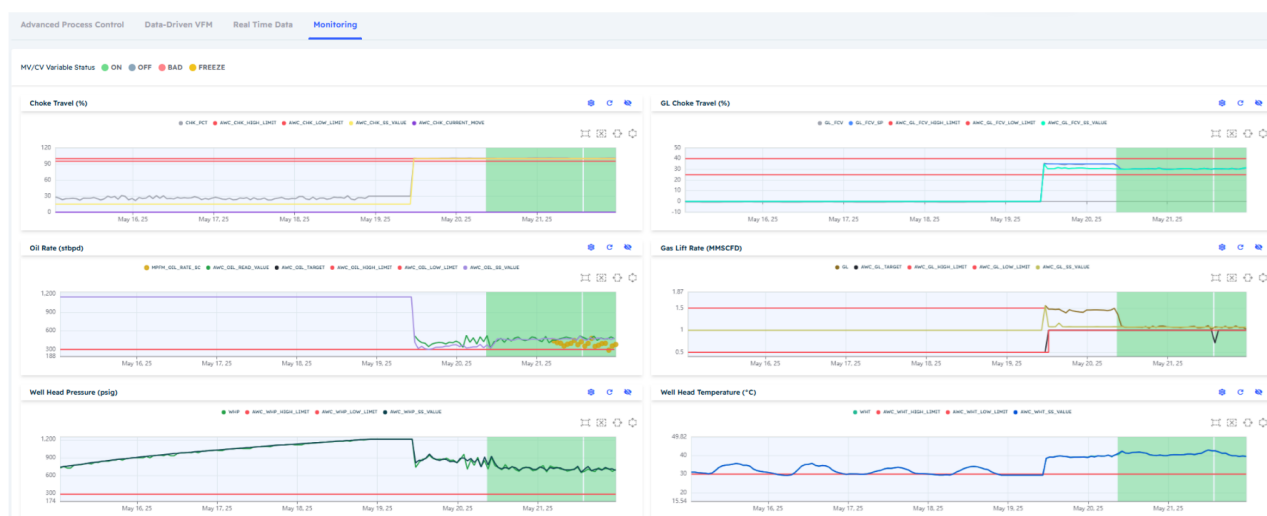


Figure 14—Impact of AWC on Well Performance Metrics

CASE 2. Another used case for Autonomous well control is to comply with well integrity autonomously and reduce well unplanned downtime or loss of well integrity that may end on additional workover cost. This can be achieved by avoiding unnecessary choke adjustments and excessive gas lift, AI APC helped prevent tubing damage and reduced the likelihood of workovers.



Figure 15—Impact of AWC on Well Integrity

CASE 3. Another application of autonomous well control pertains to ensuring compliance with operational and reservoir management guidelines, including the maintenance of bottomhole pressure within specified upper and lower thresholds, adherence to oil production targets, gas–oil ratio (GOR) constraints, and conformance with facility-design temperature and pressure limits. High and low setpoints for key process

variables are programmed within the APC controller to align with these operational requirements and constraints. Technology ensures 100% compliance and maximum energy efficiency. Figure 16 & 17 shows two tested scenarios for the same well. On the left side, a maximum production scenario was implemented allowing AWC to increase production allowing a bottom hole flowing pressure to drop up to 1500 psi (Low limit BHFP). The AWC autonomously achieved 2222 BOPD by increasing 42% gas lift consumption. Later, the low limit bottom hole pressure was increased to 2500 psi, and AWC reduced autonomously 33% of the gas lift consumption to comply with the pressure constraint to adhere to reservoir guidelines. This compliance has resulted in enhanced reservoir compliance KPI and ultimately in a higher recovery factor according to reservoir model simulation.

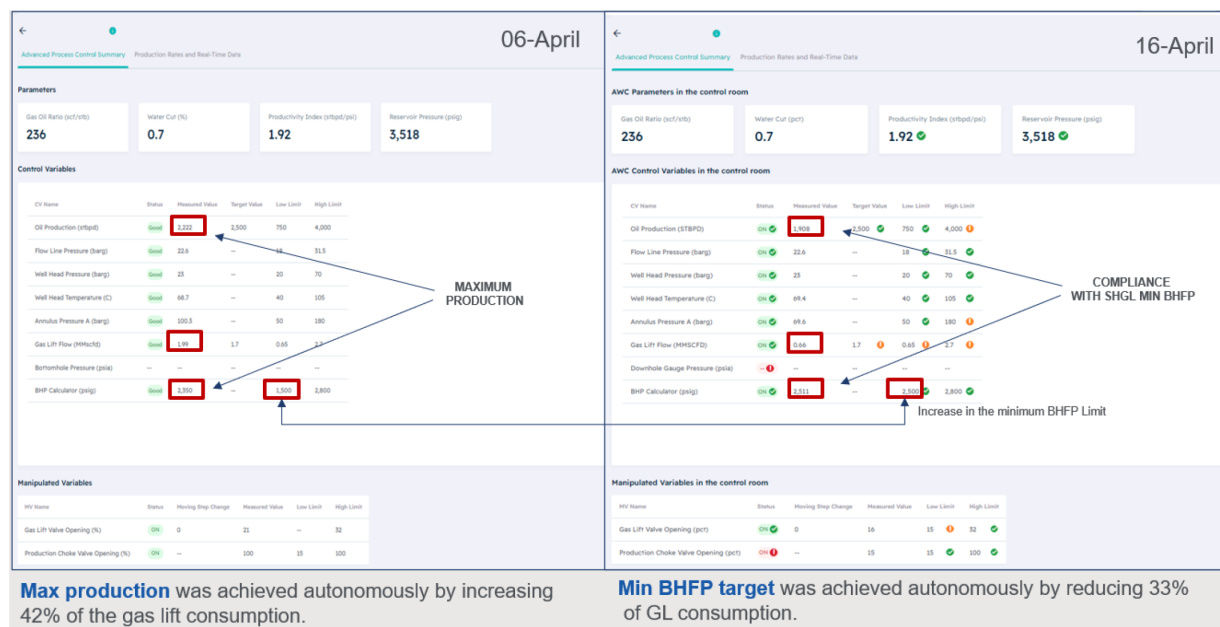


Figure 16—Effect of AWC on Reservoir Compliance Metrics

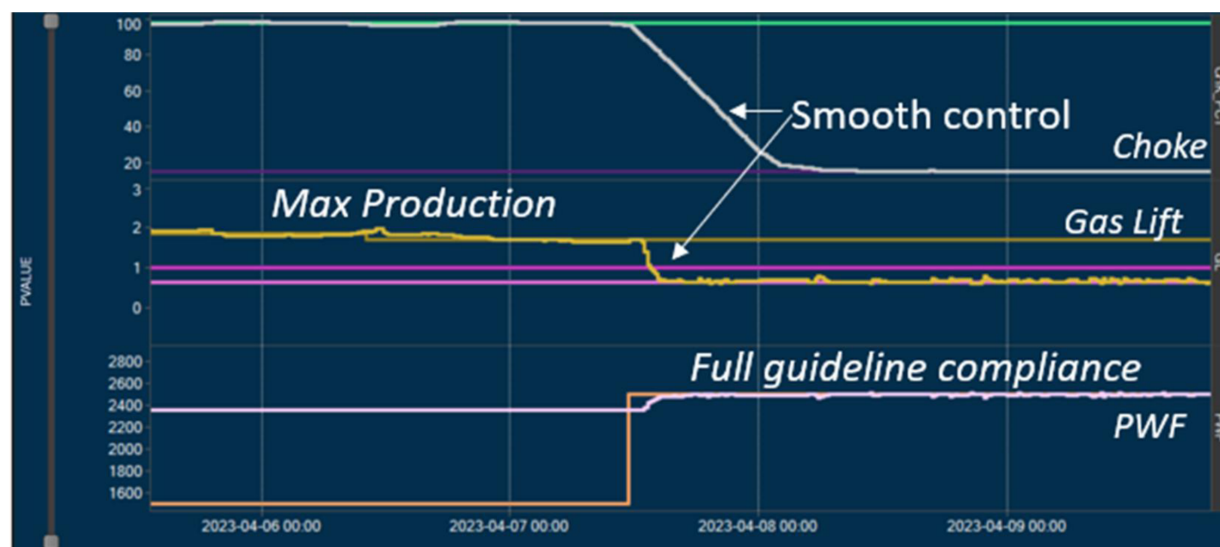


Figure 17—Effect of AWC on Reservoir Compliance Metrics

Quantitative Performance Indicators. The deployment of AI-driven autonomous gas-lift optimization across an upstream field yielded measurable improvements in production efficiency, energy utilization, and

operational stability. These outcomes were validated through quantitative performance metrics and field-level observations.

Gas Lift Injection Savings: Controller-managed injection rates were maintained near the Gas-Lift Inflection Point (GLIP), leading to a gas optimisation up to 5%-30% at well level, for most wells, with no decrease in production levels. The system dynamically adjusted injection rates based on real-time well behavior, eliminating over-injection and improving gas allocation efficiency.

Oil Production Increase: Post-deployment analysis revealed up to 1–2% uplift in average daily oil production. Specific wells demonstrated even higher gains, particularly those previously operating under suboptimal injection regimes. The optimization corrected inefficiencies in gas-lift distribution and improved reservoir drawdown, contributing directly to enhanced recovery.

Stability and Control: The system achieved smoother operation by reducing oscillations in flow rates and pressure profiles. This improved flowline management and reduced variability in separator loads, enhancing downstream process stability. The predictive control logic minimized abrupt changes in valve positions and maintained consistent operating conditions.

Unplanned Downtime Reduction: Autonomous control and predictive alerts contributed to zero downtime caused by high-pressure shutdowns. The system's ability to detect early signs of operational anomalies and initiate corrective actions proactively helped maintain continuous production and reduce intervention frequency.

These results affirm the effectiveness of AI-driven autonomous control in optimizing gas-lift operations. The integration of predictive modeling, real-time data, and automated workflows not only enhanced production and energy efficiency but also improved system reliability and reduced operational risks.

Conclusions

The implementation of AI-driven autonomous gas-lift optimization marks a significant milestone in Field Operator Company's digital transformation journey. By integrating multivariable predictive control, inferential modeling, and cloud-hosted supervisory workflows, the Well Autonomous Control System (AWC) initiative has demonstrated how intelligent automation can revolutionize well operations.

The system's ability to maintain gas-lift injections near the inflection point, dynamically adjust control parameters, and respond to real-time well behavior has led to measurable performance improvements. These include up to 5%-30% gas lift savings at well level, 1–2% uplift in oil production, enhanced operational stability, and reduced unplanned downtime. Such gains directly contribute to Field Operator Company's strategic goals of maximizing hydrocarbon recovery while minimizing energy consumption and operational risk.

Beyond technical performance, the initiative has catalyzed a broader digital transformation. The deployment of integrated dashboards, automated workflows, and centralized model management has streamlined collaboration across field operations, petroleum engineering, and digital teams. This shift from manual, reactive workflows to intelligent, closed-loop optimization has improved agility, reduced latency in decision-making, and enhanced cross-functional visibility.

The study validates that AI-enabled autonomous optimization is not only feasible but essential for future-ready oilfield management. As computational infrastructure and data acquisition technologies continue to evolve, the scalability and impact of such systems will expand. Continued collaboration between industry practitioners, technology partners, and academic researchers will be key to refining these models and extending their application across diverse reservoir and production environments ultimately enabling sustainable production enhancement through digital transformation.

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Nomenclature

AI	: Artificial Intelligence
ANN	: Artificial Neural Network
APC	: Advanced Process Control
API	: Application Programming Interface
AWC	: Well Autonomous Control System
BHFP	: Bottom Hole Flowing Pressure
BHPIM	: Bottom Hole Pressure Inferential Model
BSW	: Basic Sediment & Water
CSV	: Comma Separated Values
CV	: Controlled Variables
DCS	: Distributed Control System
DV	: Disturbance Variables
GA	: Genetic Algorithm
GL	: Gas Lift
GLIP	: Gas-Lift Inflection Point
GLPRIM	: Gas Lift Production Ratio Inferential Model
GOR	: Gas Oil Ratio
IPR	: Inflow Performance Relationships
I&C	: Instrumentation & Control
IT	: Information technology
KPI	: Key Performance Indicator
MAPE	: Median Mean Absolute Percentage Error
MAASP	: Maximum Allowable Annulus Surface Pressure
MPC	: Multivariable Predictive Control
MPFM	: Multi- Phase Flow Meter
MV	: Manipulated Variables
OT	: Operator Technology
PI	: Productivity Index
PRIM	: Production Rate Inferential Model
PWF	: Flowing Bottom Hole Pressure
RSM	: Response Surface Methodology
VFM	: Virtual Flow Meter
WC	: Water Cut
WHP	: Well Head Pressure

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